

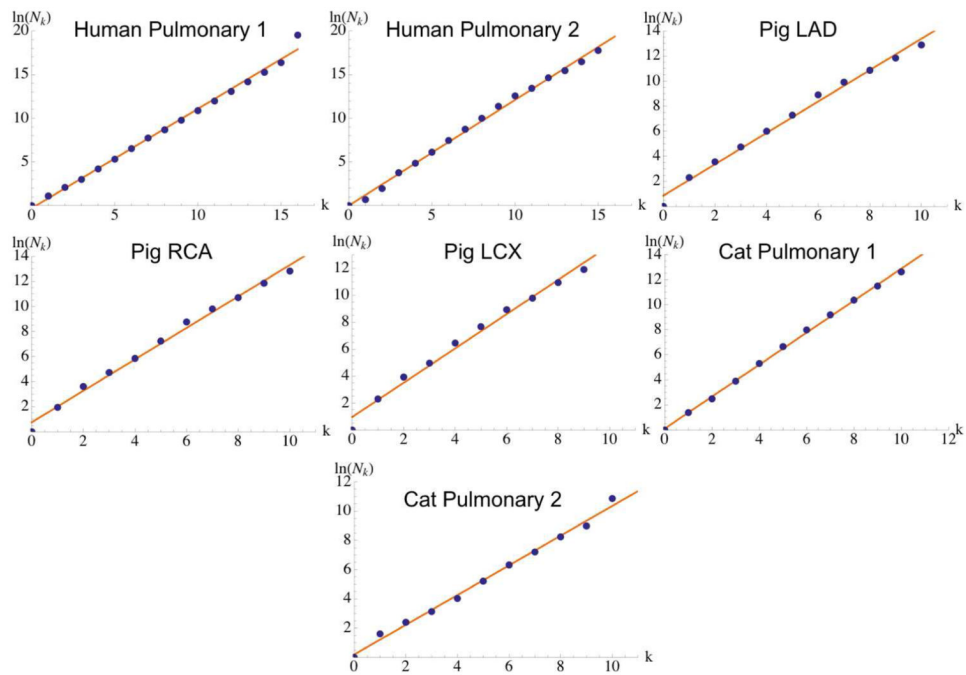


Figure 1.

Plot of the logarithm of the number of vessels, N_k versus the branching order, k , as determined by the Horton-Strahler method[34, 87]. Data are from [37, 99, 100, 45, 35] from 8 healthy sections (e.g., left anterior descending artery, pulmonary) of vascular systems in pig, cat, and human. In all cases, the correlations are excellent and corresponding branching ratios are near $n = 3$. See Fig. 6 for estimates of scaling exponents based on best fit regression lines as well as p-values, 95% confidence intervals, and r^2 values.

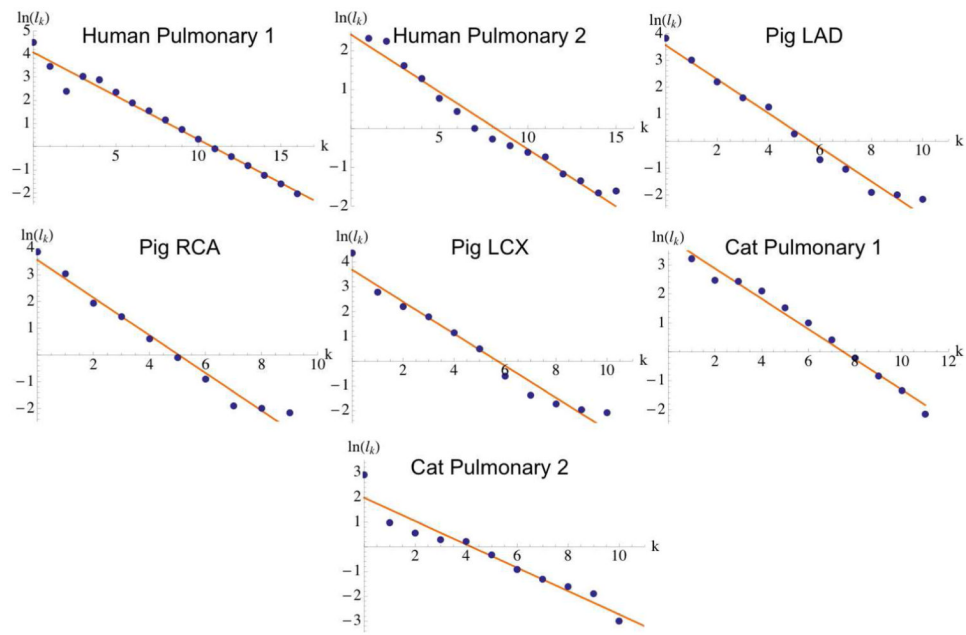


Figure 2.

Plot of the logarithm of the length of vessels, l_k , versus the branching order, k , as determined by the Strahler method. Data are from [37, 99, 100, 45, 35, 56, 55] from 7 healthy sections (e.g., left anterior descending artery, pulmonary) of vascular systems in pig, cat, and human. For healthy tissue, the correlations are excellent and correspond to scaling exponents that on average are close to the value of $b = 1/2$ (see Fig. 6), somewhat higher than predicted. See Fig. 6 for estimates of scaling exponents based on best fit regression lines as well as p-values, 95% confidence intervals, and r^2 values.

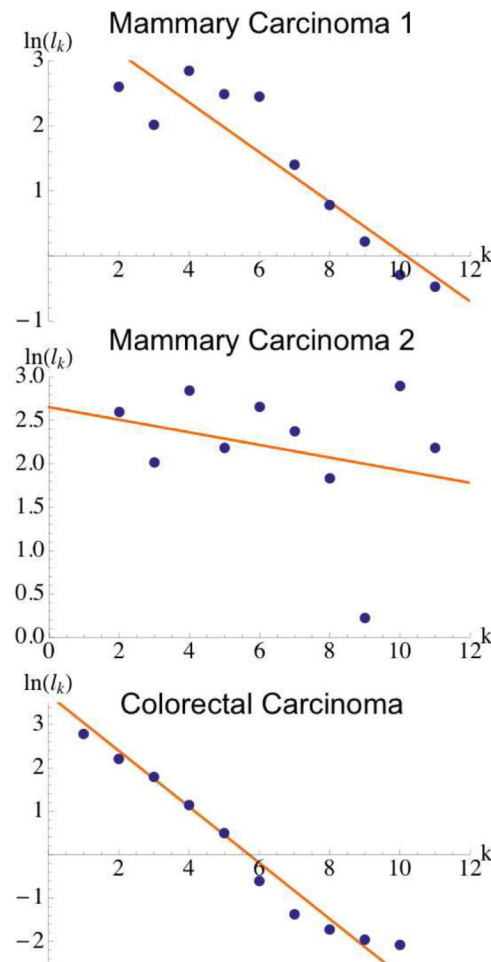


Figure 3.

Plot of the logarithm of the length of vessels, l_k , versus the branching order, k , as determined by the Strahler method. Data are from tumors (mammary and colorectal carcinomas (A91 and A97)) in rat and human. For tumors, the correlations are much weaker and the scaling exponents are significantly shallower than for healthy tissue shown in Fig. 2. For both healthy tissue and tumors, a self-similar fractal network is a reasonable approximation for the vascular systems, but for tumors there is still some significant variation to explain. See Fig. 6 for estimates of scaling exponents based on best fit regression lines as well as p-values, 95% confidence intervals, and r^2 values.

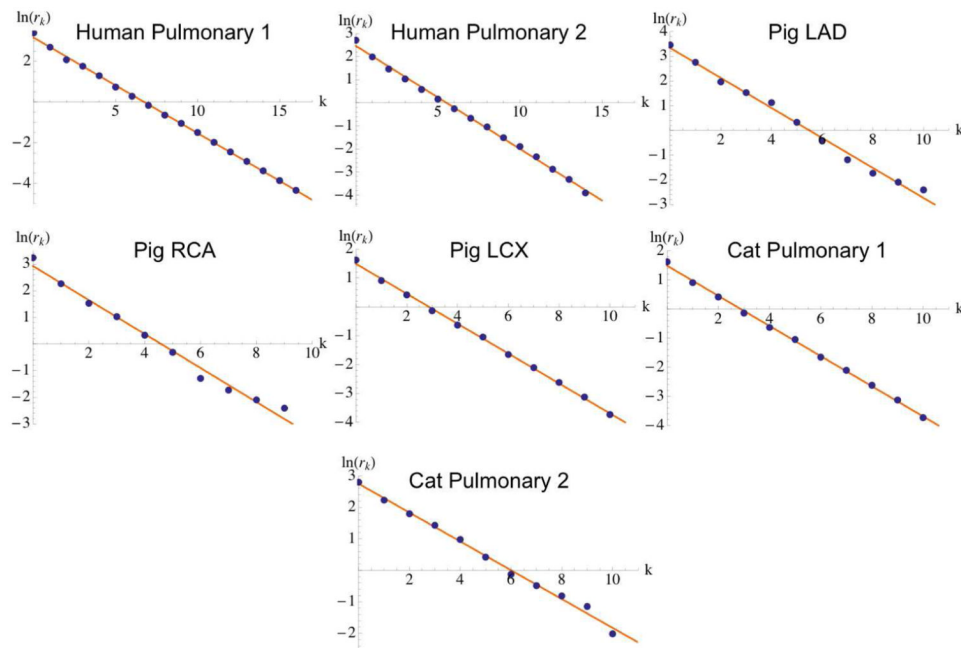


Figure 4.

Plot of the logarithm of the radii of vessels, r_k , versus the branching order, k , as determined by the Strahler method. Data are from [37, 99, 100, 45, 35, 56, 55] from 7 healthy sections (e.g., left anterior descending artery, pulmonary) vascular systems in pig, cat, and human. For healthy tissue, the correlations are excellent and correspond to scaling exponents that on average are close to the predicted value of $a = 1/2$ for larger vessels (see Fig. 6). Also see Fig. 6 for estimates of scaling exponents based on best fit regression lines as well as p-values, 95% confidence intervals, and r^2 values.

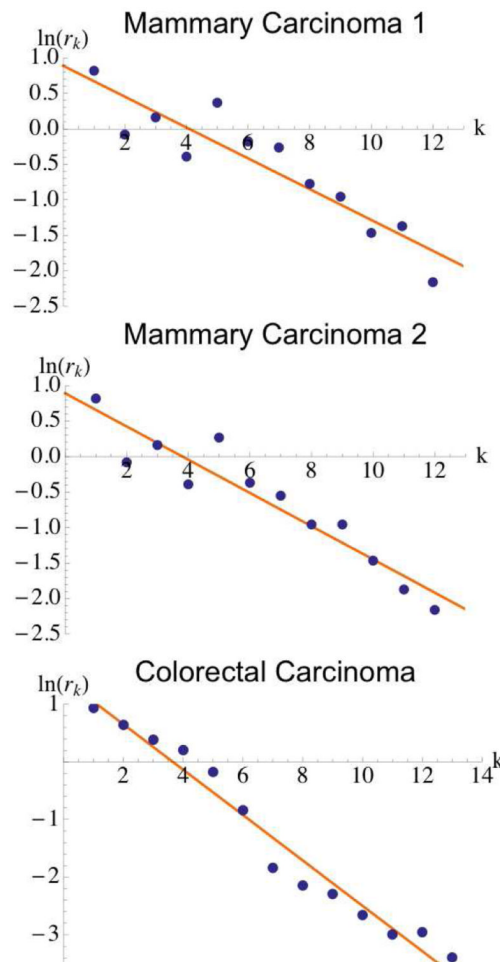


Figure 5.

Plot of the logarithm of the radii of vessels, r_k , versus the branching order, k , as determined by the Strahler method. Data are from tumors (mammary and colorectal carcinomas (A91 and A97)) in rat and human. For tumors, the correlations are weaker and the scaling exponents are significantly shallower than for healthy tissue shown in Fig. 4. For both healthy tissue and tumors, a self-similar fractal network is a reasonable approximation for the vascular systems, but for tumors there still remains significant variation to explain. See Fig. 6 for estimates of scaling exponents based on best fit regression lines as well as p-values, 95% confidence intervals, and r^2 values.

vascular system	branching ratio, n	scaling exponent radius, a (n=3)	r ²	p-value	scaling exponent length, b (n=3)	r ²	p-value
Human Pulmonary 1	3.12±1.05	0.43±0.01	1	<0.0001	0.34±0.03	0.98	<0.0001
Human Pulmonary 2	3.35±1.04	0.41±0.01	1	<0.0001	0.27±0.03	0.97	<0.0001
Pig LAD	3.50±1.09	0.55±0.04	0.99	<0.0001	0.58±0.06	0.98	<0.0001
Pig RCA	3.51±1.08	0.58±0.06	0.99	<0.0001	0.64±0.08	0.98	<0.0001
Pig LCX	3.57±1.12	0.52±0.04	0.99	<0.0001	0.59±0.08	0.97	<0.0001
Cat Pulmonary 1	3.58±1.04	0.47±0.01	1	<0.0001	0.48±0.05	0.98	<0.0001
Cat Pulmonary 2	2.76±1.06	0.42±0.02	0.99	<0.0001	0.43±0.09	0.93	<0.0001
Mean of Healthy	3.34	0.49	0.99	<0.0001	0.48	0.97	<0.0001
Mammary Carcinoma 1	N/A	0.20±0.06	0.84	<0.0001	0.35±0.12	0.84	0.00018
Mammary Carcinoma 2	N/A	0.21±0.05	0.89	<0.0001	0.07±0.18	0.08	0.42989
Colorectal Carcinoma	N/A	0.36±0.04	0.97	<0.0001	0.24±0.13	0.67	0.00209
Mean Tumor	N/A	0.25	0.90	<0.0001	0.22	0.53	0.14405

Figure 6.

Table of calculated scaling parameters and correlation coefficients for vascular systems based on data in Figs. 1–5. Average values for healthy tissue and tumors are shown in bold. Tumor vasculature exhibits higher variation (with p-values for length scaling exponents all being > 0.0001), possibly corresponding to noise in gene expression or chromosome segregation, and also shallower scaling exponents, representing slower scaling and possible representing a breakdown in vessel hierarchy.

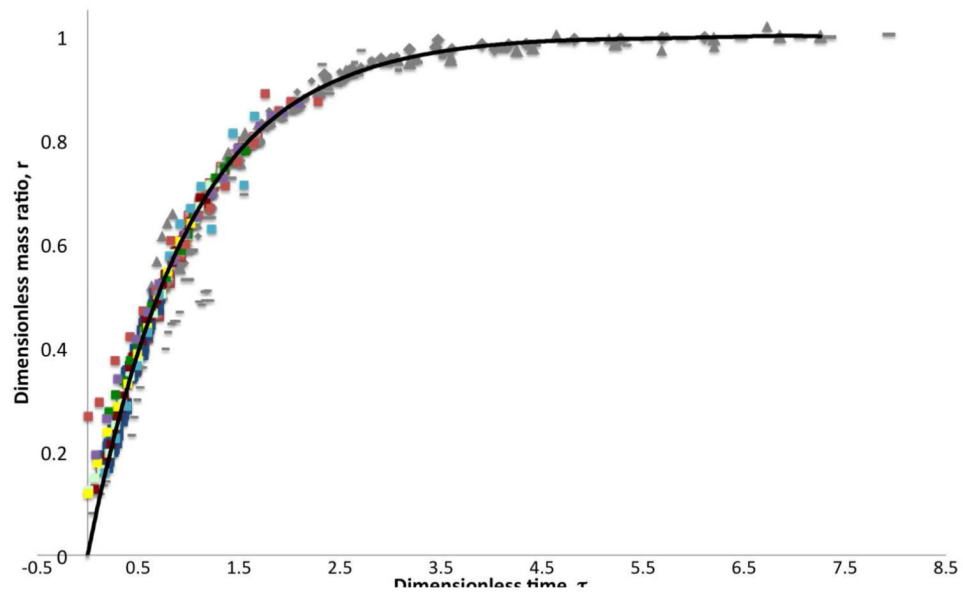


Figure 7.

Plots of dimensionless ratio versus dimensionless time as defined by Eqs. (29)-(30). Data in grey are for ontogenetic growth from 13 species of animals (see [98] for original data sources), ranging from guppy to cod to guinea pig, and data in color with square symbols are for tumor growth trajectories for C3H mammary carcinoma (dark blue), EMT6 mammary carcinoma (dark red), KHJJ mammary carcinoma (light green), NCTC (dark green), Flank (yellow), Primary fibroadenoma (red), Primary Osteosarcoma (light blue), and Walker Carcinoma (purple) tumor types (see [31, 92] for original data references). This plot shows the extent to which all the growth data can be described by a single universal curve [98, 24], and thus, the extent to which tumor growth is really analogous to ontogenetic growth from birth to adult. Fitted parameter values are taken from the original references.

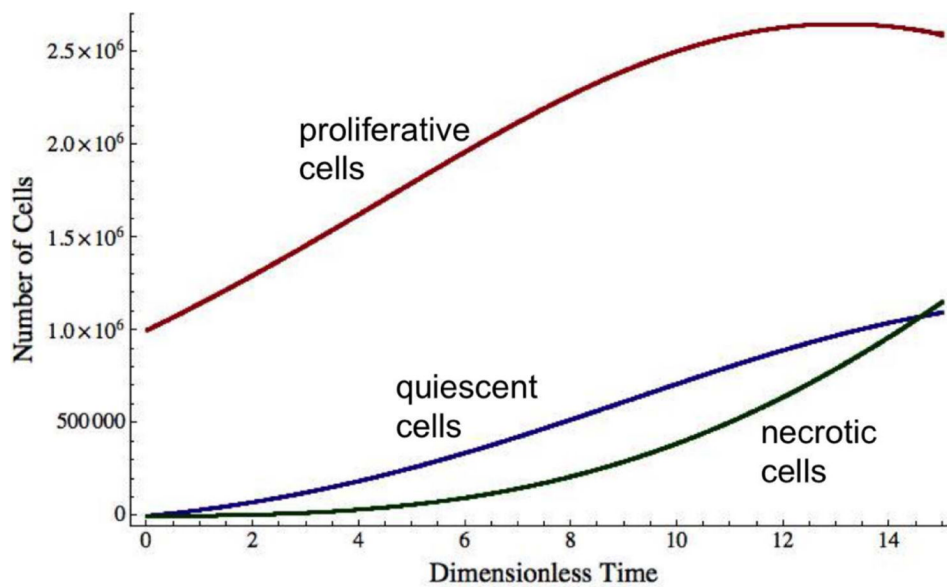


Figure 8.

Plot of numerical solutions to find model predictions for number of proliferative (red), quiescent (blue), and necrotic (green) cells versus time within a model tumor. Parameter

values are $A = B_0 m_c / E_c = 0.305$ (1/day), $\beta = \frac{1}{2a+b} = 1$, $F = B_m / E_c = 0.14$ (1/day), $\lambda = 0$, $\alpha' = 1.4$, and $\nu = 0.01$ (1/day).

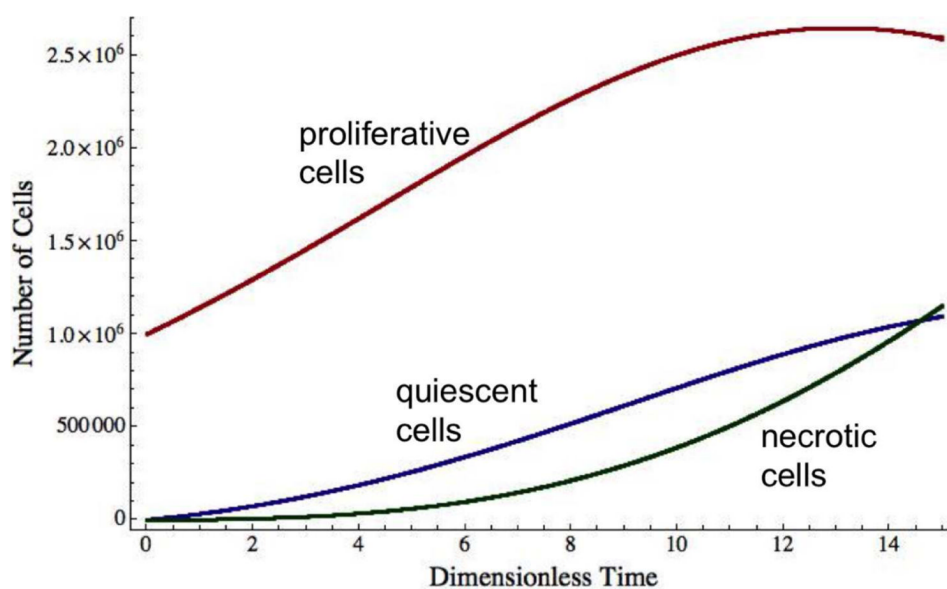


Figure 9. Timing sequences for drug delivery for the three scenarios described in the text. Parameters for the drugs are $d = 100$ units per volume, threshold for cell death is 10 units per volume, and $\Delta t = 7 \times 10^{-4}$ days is the time step.

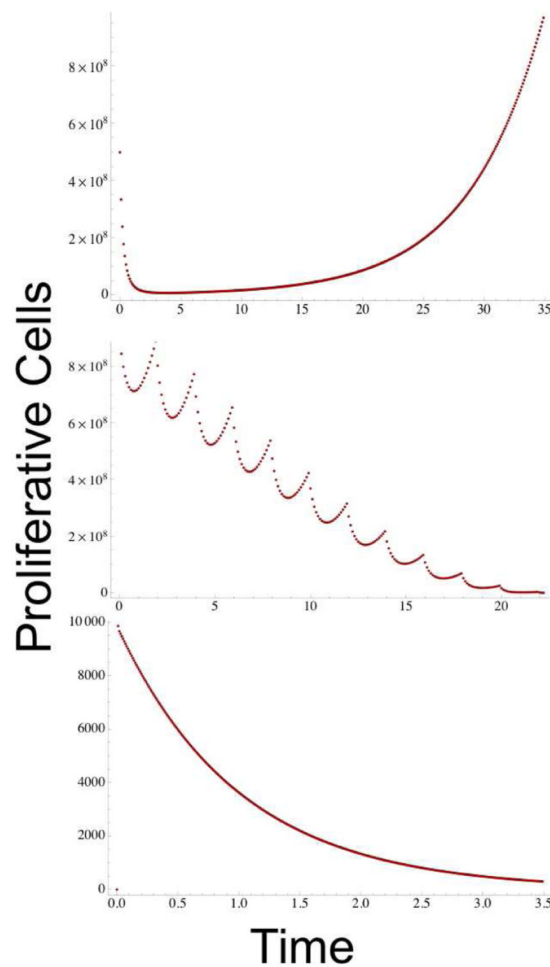


Figure 10.

Plots for resultant trajectories of proliferative cell numbers in response to the corresponding (by row from top to bottom) sequence of drug treatments in Fig. 9. These plots are model predictions based on numerical solutions obtained by evolving the equations through time.

Parameter values are $A = B_0 m_c / E_c = 0.305$ (1/day), $\beta = \frac{1}{2a+b} = 1$, $F = B_m / E_c = 0.14$ (1/day), $\lambda = 0$, $\alpha' = 1.4$, and $\nu = 0.01$ (1/day). Parameters for pressure are, $\kappa = 36.823$ (taken from [5]), $p_n = 10$ mm Hg, and $p_o = 3$ mm Hg. Power thresholds for transition to quiescence and necrosis are given by $B_{c,q} = 0.001$ J/day and $B_{c,n} = 0.0007$ J/day respectively.

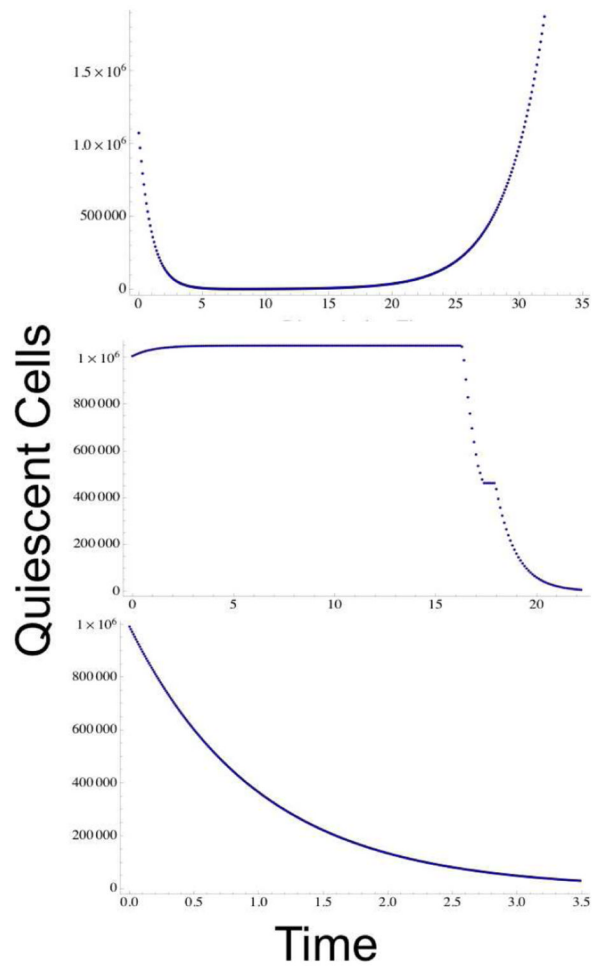


Figure 11.

Plots for resultant trajectories of quiescent cell numbers in response to the corresponding (by row from top to bottom) sequence of drug treatments in Fig. 9. These plots are model predictions based on numerical solutions obtained by evolving the equations through time. Parameter values are the same as are given in Fig. 10.

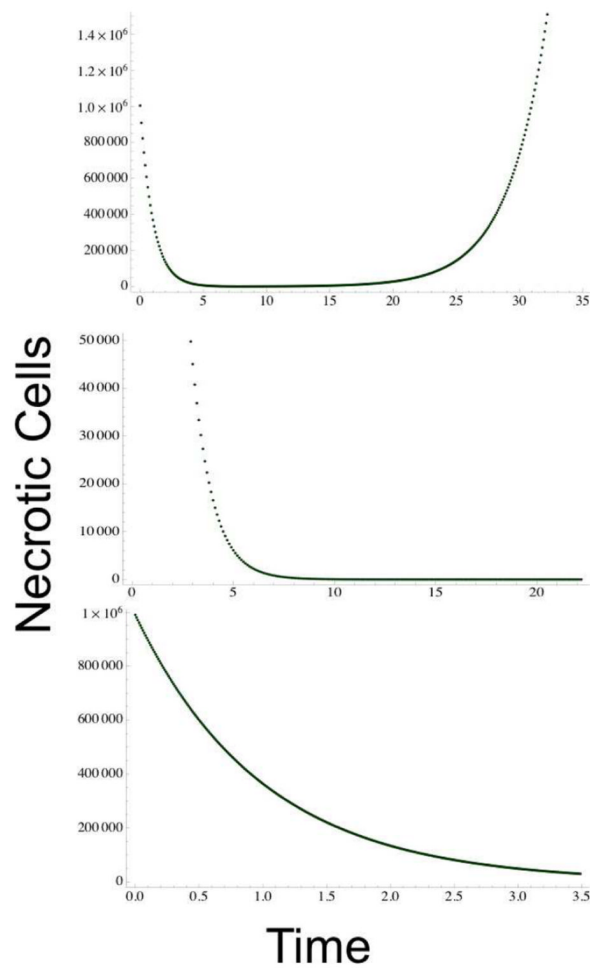


Figure 12.

Plots for resultant trajectories of necrotic cell numbers in response to the corresponding (by row from top to bottom) sequence of drug treatments in Fig. 9. These plots are model predictions based on numerical solutions obtained by evolving the equations through time. Parameter values are the same as are given in Fig. 10.